STA302_Project

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Background

This project investigates factors associated with apartment building evaluation scores in Toronto rental properties using data from the Toronto Open Data Apartment Building Evaluation dataset.

The study focuses on how building characteristics and maintenance-related maintenance factors are associated with current apartment building evaluation scores.

Research Question

What building characteristics and maintenance-related factors are associated with higher apartment building evaluation scores in Toronto rental properties?

Literature Review

Previous studies suggest that housing quality and residential satisfaction are associated with building age, maintenance conditions, and structural characteristics.

Kutty (1999) found that housing adequacy is strongly associated with building characteristics and housing conditions. Zhong and Gou (2023) used RentSafeTO data to examine apartment building quality in Toronto and identified relationships between building conditions and residential quality.

Data Description

The dataset used in this project is the Apartment Building Evaluation dataset from Toronto Open Data.

The cleaned dataset contains 5304 apartment building observations after removing rows with missing values.

Data Cleaning

The raw dataset was imported and cleaned before analysis. Relevant predictors and the response variable were selected based on findings from the literature review.

Rows containing missing values were removed using complete-case analysis. Categorical variables were converted into factor format for regression analysis.

```
data <- read.csv("data/FinalProject_dataset.csv")
dim(data)
```

```
## [1] 5340   71
```

```
names(data)
```

```
##  [1] "X_id"                           "RSN"
##  [3] "YEAR.REGISTERED"                "YEAR.BUILT"
##  [5] "YEAR.EVALUATED"                 "PROPERTY.TYPE"
##  [7] "WARD"                           "WARDNAME"
##  [9] "SITE.ADDRESS"                   "CONFIRMED.STOREYS"
## [11] "CONFIRMED.UNITS"                "EVALUATION.COMPLETED.ON"
## [13] "CURRENT.BUILDING.EVAL.SCORE"    "PROACTIVE.BUILDING.SCORE"
## [15] "CURRENT.REACTIVE.SCORE"         "NO.OF.AREAS.EVALUATED"
## [17] "NUMBERING.OF.PROPERTY"          "EXTERIOR.GROUNDS"
## [19] "FENCING"                        "RETAINING.WALLS"
## [21] "CATCH.BASINS...STORM.DRAINAGE"  "BUILDING.EXTERIOR"
## [23] "BALCONY.GUARDS"                 "WINDOWS"
## [25] "EXT..RECEPTACLE.STORAGE.AREA"   "EXTERIOR.WALKWAYS"
## [27] "CLOTHING.DROP.BOXES"            "ACCESSORY.BUILDINGS"
## [29] "INTERCOM"                       "EMERGENCY.CONTACT.SIGN"
## [31] "LOBBY...WALLS.AND.CEILING"      "LOBBY.FLOORS"
## [33] "LAUNDRY.ROOM"                   "INT..RECEPTACLE.STORAGE.AREA"
## [35] "MAIL.RECEPTACLES"               "EXTERIOR.DOORS"
## [37] "STORAGE.AREAS.LOCKERS.MAINT."   "POOLS"
## [39] "OTHER.AMENITIES"                "PARKING.AREAS"
## [41] "ABANDONED.EQUIP..DERELICT.VEH." "GARBAGE.COMPACTOR.ROOM"
## [43] "ELEVATOR.MAINTENANCE"           "ELEVATOR.COSMETICS"
## [45] "INT..HALLWAY...WALLS...CEILING" "INTERIOR.HALLWAY.FLOORS"
## [47] "INT..LOBBY...HALLWAY.LIGHTING"  "COMMON.AREA.VENTILATION"
## [49] "ELECTRICAL.SERVICES...OUTLETS"  "CHUTE.ROOMS...MAINTENANCE"
## [51] "STAIRWELL...WALLS.AND.CEILING"  "STAIRWELL...LANDING.AND.STEPS"
## [53] "STAIRWELL.LIGHTING"             "INT..HANDRAIL...GUARD...SAFETY"
## [55] "INT..HANDRAIL...GUARD...MAINT." "GRAFFITI"
## [57] "BUILDING.CLEANLINESS"           "COMMON.AREA.PESTS"
## [59] "TENANT.NOTIFICATION.BOARD"      "PEST.CONTROL.LOG"
## [61] "MAINTENANCE.LOG"                "CLEANING.LOG"
## [63] "VITAL.SERVICE.PLAN"             "ELECTRICAL.SAFETY.PLAN"
## [65] "STATE.OF.GOOD.REPAIR.PLAN"      "TENANT.SERVICE.REQUEST.LOG"
## [67] "GRID"                           "LATITUDE"
## [69] "LONGITUDE"                      "X"
## [71] "Y"
```

```
str(data)
```

```
## 'data.frame':	5340 obs. of  71 variables:
##  $ X_id                          : int  1 2 3 4 5 6 7 8 9 10 ...
##  $ RSN                           : int  4154249 4639099 4155151 4155151 4153401 41534
01 4153400 4152941 4155081 4155081 ...
##  $ YEAR.REGISTERED               : int  2017 2019 2017 2017 2017 2017 2017 2017 2017
2017 ...
##  $ YEAR.BUILT                    : int  1966 2018 1940 1940 1979 1979 1992 1958 1996
1996 ...
##  $ YEAR.EVALUATED                : int  2025 2024 2025 2023 2023 2025 2024 2024 2023
2025 ...
##  $ PROPERTY.TYPE                 : chr  "PRIVATE" "PRIVATE" "PRIVATE" "PRIVATE" ...
##  $ WARD                          : int  7 5 5 5 10 10 10 4 5 5 ...
##  $ WARDNAME                      : chr  "Humber River-Black Creek" "York South-Westo
n" "York South-Weston" "York South-Weston" ...
##  $ SITE.ADDRESS                  : chr  "10 JAYZEL DR" "22 JOHN ST" "50 JOHN ST" "50
JOHN ST" ...
##  $ CONFIRMED.STOREYS             : int  7 31 3 3 7 7 4 4 11 11 ...
##  $ CONFIRMED.UNITS               : int  90 369 14 14 180 180 26 42 132 132 ...
##  $ EVALUATION.COMPLETED.ON       : chr  "2025-10-20" "2024-06-20" "2025-11-14" "2023-
08-25" ...
##  $ CURRENT.BUILDING.EVAL.SCORE   : num  93 94 93 94 91 94 73 91 87 95 ...
##  $ PROACTIVE.BUILDING.SCORE      : num  93 94 93 94 91 94 74 91 87 95 ...
##  $ CURRENT.REACTIVE.SCORE        : int  0 0 0 0 0 0 -1 0 0 0 ...
##  $ NO.OF.AREAS.EVALUATED         : int  43 43 38 34 43 44 44 40 43 45 ...
##  $ NUMBERING.OF.PROPERTY         : int  3 3 3 3 3 3 3 3 3 3 ...
##  $ EXTERIOR.GROUNDS              : chr  "3" "2" "3" "3" ...
##  $ FENCING                       : chr  "2" "N/A" "3" "2" ...
##  $ RETAINING.WALLS               : chr  "2" "3" "3" "N/A" ...
##  $ CATCH.BASINS...STORM.DRAINAGE : chr  "3" "3" "3" "N/A" ...
##  $ BUILDING.EXTERIOR             : chr  "2" "3" "2" "2" ...
##  $ BALCONY.GUARDS                : chr  "3" "3" "N/A" "N/A" ...
##  $ WINDOWS                       : chr  "3" "3" "3" "3" ...
##  $ EXT..RECEPTACLE.STORAGE.AREA  : chr  "3" "N/A" "3" "3" ...
##  $ EXTERIOR.WALKWAYS             : chr  "2" "3" "3" "3" ...
##  $ CLOTHING.DROP.BOXES           : chr  "N/A" "N/A" "N/A" "N/A" ...
##  $ ACCESSORY.BUILDINGS           : chr  "N/A" "N/A" "N/A" "N/A" ...
##  $ INTERCOM                      : chr  "3" "3" "3" "N/A" ...
##  $ EMERGENCY.CONTACT.SIGN        : int  3 1 3 3 3 3 3 3 3 3 ...
##  $ LOBBY...WALLS.AND.CEILING     : chr  "3" "3" "3" "3" ...
##  $ LOBBY.FLOORS                  : chr  "3" "3" "3" "3" ...
##  $ LAUNDRY.ROOM                  : chr  "3" "N/A" "3" "2" ...
##  $ INT..RECEPTACLE.STORAGE.AREA  : chr  "N/A" "3" "N/A" "N/A" ...
##  $ MAIL.RECEPTACLES              : chr  "3" "3" "3" "3" ...
##  $ EXTERIOR.DOORS                : chr  "2" "3" "3" "3" ...
##  $ STORAGE.AREAS.LOCKERS.MAINT.  : chr  "N/A" "3" "N/A" "N/A" ...
##  $ POOLS                         : chr  "N/A" "N/A" "N/A" "N/A" ...
##  $ OTHER.AMENITIES               : chr  "N/A" "3" "N/A" "N/A" ...
##  $ PARKING.AREAS                 : chr  "2" "N/A" "3" "2" ...
##  $ ABANDONED.EQUIP..DERELICT.VEH.: int  3 3 3 3 3 3 1 3 3 3 ...
##  $ GARBAGE.COMPACTOR.ROOM        : chr  "3" "3" "N/A" "N/A" ...
##  $ ELEVATOR.MAINTENANCE          : chr  "3" "3" "N/A" "N/A" ...
##  $ ELEVATOR.COSMETICS            : chr  "3" "3" "N/A" "N/A" ...
```

```
##  $ INT..HALLWAY...WALLS...CEILING: chr  "3" "2" "3" "3" ...
##  $ INTERIOR.HALLWAY.FLOORS       : chr  "N/A" "2" "3" "3" ...
##  $ INT..LOBBY...HALLWAY.LIGHTING : chr  "3" "3" "3" "3" ...
##  $ COMMON.AREA.VENTILATION       : chr  "2" "3" "N/A" "N/A" ...
##  $ ELECTRICAL.SERVICES...OUTLETS : chr  "3" "3" "3" "N/A" ...
##  $ CHUTE.ROOMS...MAINTENANCE     : chr  "3" "2" "N/A" "N/A" ...
##  $ STAIRWELL...WALLS.AND.CEILING : chr  "3" "3" "3" "3" ...
##  $ STAIRWELL...LANDING.AND.STEPS : chr  "3" "3" "2" "2" ...
##  $ STAIRWELL.LIGHTING            : chr  "3" "3" "3" "3" ...
##  $ INT..HANDRAIL...GUARD...SAFETY: chr  "3" "3" "1" "3" ...
##  $ INT..HANDRAIL...GUARD...MAINT.: chr  "3" "3" "2" "3" ...
##  $ GRAFFITI                      : int  2 3 3 3 2 3 1 3 3 3 ...
##  $ BUILDING.CLEANLINESS          : int  3 2 3 3 3 1 2 3 2 3 ...
##  $ COMMON.AREA.PESTS             : int  3 3 3 3 2 3 3 3 3 3 ...
##  $ TENANT.NOTIFICATION.BOARD     : int  3 2 3 3 3 3 1 3 2 3 ...
##  $ PEST.CONTROL.LOG              : int  3 3 3 3 3 3 3 3 2 3 ...
##  $ MAINTENANCE.LOG               : int  3 3 3 3 3 3 3 3 3 3 ...
##  $ CLEANING.LOG                  : int  3 3 3 3 2 3 1 3 3 3 ...
##  $ VITAL.SERVICE.PLAN            : int  3 3 3 3 3 3 3 3 3 3 ...
##  $ ELECTRICAL.SAFETY.PLAN        : int  3 3 2 3 3 3 3 3 2 3 ...
##  $ STATE.OF.GOOD.REPAIR.PLAN     : int  3 2 3 3 3 3 3 3 3 3 ...
##  $ TENANT.SERVICE.REQUEST.LOG    : int  3 3 3 3 3 3 3 3 3 3 ...
##  $ GRID                          : chr  "W0727" "W0524" "W0525" "W0525" ...
##  $ LATITUDE                      : num  43.7 NA 43.7 43.7 43.7 ...
##  $ LONGITUDE                     : num  -79.5 NA -79.5 -79.5 -79.4 ...
##  $ X                             : num  300804 303361 303461 303461 313521 ...
##  $ Y                             : num  4845265 4839991 4840158 4840158 4834252 ...
```

```
data_subset <- data[, c(
  "CURRENT.BUILDING.EVAL.SCORE",
  "PROPERTY.TYPE",
  "YEAR.BUILT",
  "CONFIRMED.UNITS",
  "CONFIRMED.STOREYS",
  "BUILDING.CLEANLINESS",
  "BUILDING.EXTERIOR",
  "COMMON.AREA.PESTS"
)]

data_subset[data_subset == "N/A"] <- NA
clean_data <- na.omit(data_subset)
dim(clean_data)
```

```
## [1] 5304    8
```

```
clean_data$PROPERTY.TYPE <- as.factor(clean_data$PROPERTY.TYPE)
clean_data$BUILDING.EXTERIOR <- as.numeric(clean_data$BUILDING.EXTERIOR)
```

```
# FinalProject_dataset output
write.csv(clean_data,
          "output/cleaned_apartment_building_data.csv",
          row.names = FALSE)
```

Linear Regression Analysis and Residual Analysis

A preliminary multiple linear regression model was fitted using predictors identified in the literature review.

Residual diagnostic plots were used to assess the assumptions of linear regression, including approximate normality and linearity.

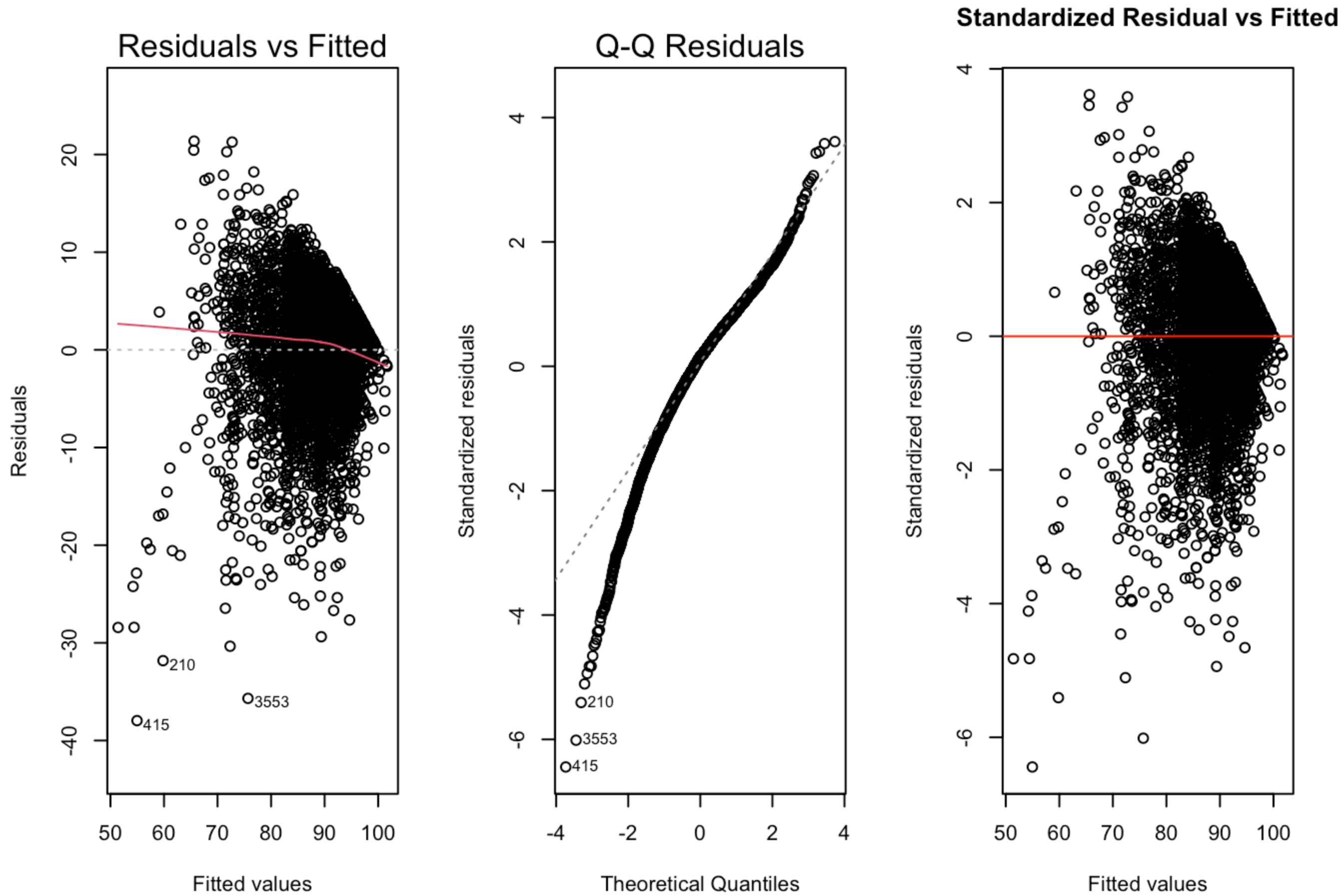
```
# Linear Regression Model
model <- lm( CURRENT.BUILDING.EVAL.SCORE ~ PROPERTY.TYPE + YEAR.BUILT +
    CONFIRMED.UNITS + CONFIRMED.STOREYS + BUILDING.CLEANLINESS +
    BUILDING.EXTERIOR + COMMON.AREA.PESTS,
  data = clean_data )

summary(model)
```

```
##
## Call:
## lm(formula = CURRENT.BUILDING.EVAL.SCORE ~ PROPERTY.TYPE + YEAR.BUILT +
##     CONFIRMED.UNITS + CONFIRMED.STOREYS + BUILDING.CLEANLINESS +
##     BUILDING.EXTERIOR + COMMON.AREA.PESTS, data = clean_data)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -37.962  -3.080   0.878   3.913  21.377
##
## Coefficients:
##                              Estimate Std. Error t value Pr(>|t|)
## (Intercept)                -26.655106   9.424639  -2.828 0.004698 **
## PROPERTY.TYPESOCIAL HOUSING   0.340383   0.350078   0.972 0.330942
## PROPERTY.TYPETCHC            1.029439   0.305614   3.368 0.000761 ***
## YEAR.BUILT                   0.037806   0.004841   7.810 6.87e-15 ***
## CONFIRMED.UNITS              0.004355   0.001710   2.547 0.010905 *
## CONFIRMED.STOREYS            0.087001   0.025665   3.390 0.000705 ***
## BUILDING.CLEANLINESS         6.248578   0.142037  43.993  < 2e-16 ***
## BUILDING.EXTERIOR            3.719292   0.128763  28.885  < 2e-16 ***
## COMMON.AREA.PESTS            5.126644   0.279272  18.357  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 5.948 on 5295 degrees of freedom
## Multiple R-squared:  0.5093, Adjusted R-squared:  0.5086
## F-statistic:   687 on 8 and 5295 DF,  p-value: < 2.2e-16
```

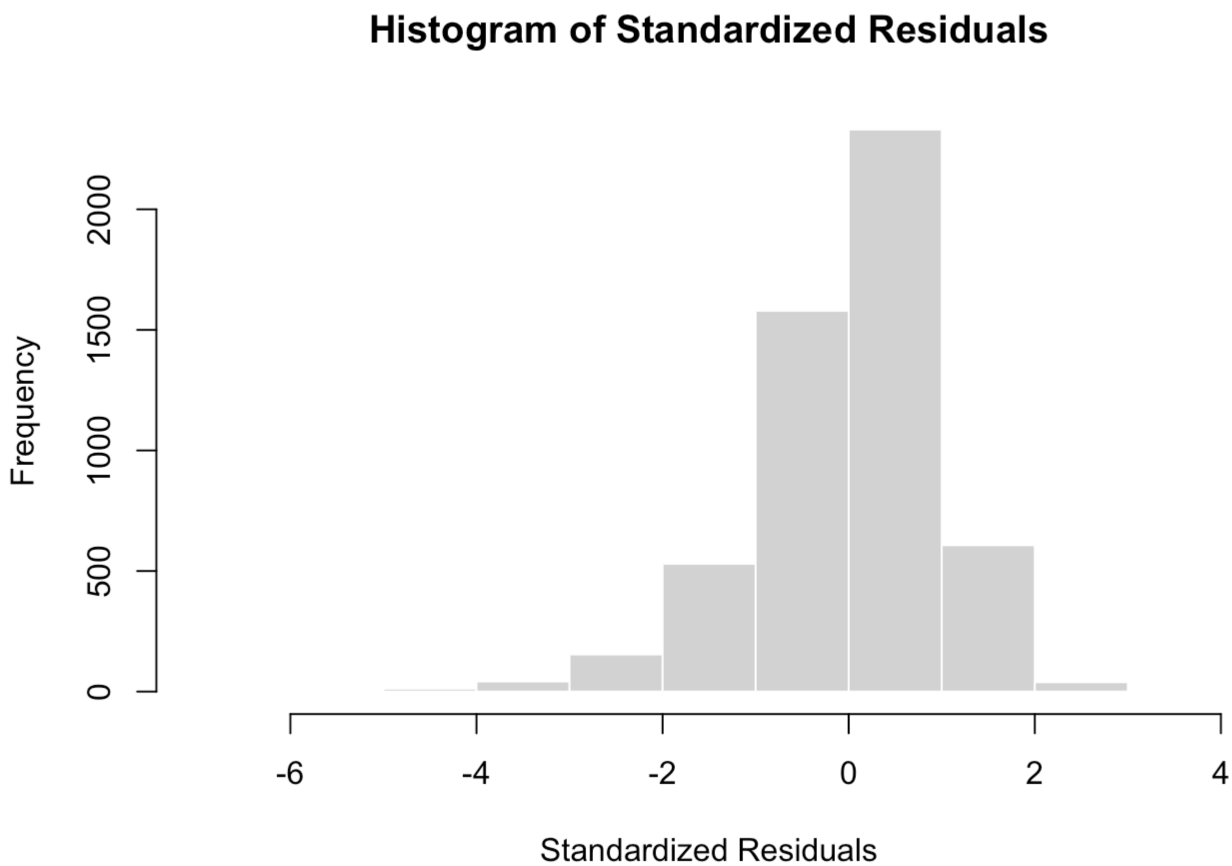
```
confint(model)
```

```
##                                       2.5 %       97.5 %
## (Intercept)                  -45.131283036 -8.178929938
## PROPERTY.TYPESOCIAL HOUSING   -0.345913967  1.026680689
## PROPERTY.TYPETCHC              0.430309844  1.628569039
## YEAR.BUILT                     0.028316114  0.047296775
## CONFIRMED.UNITS                0.001002519  0.007707659
## CONFIRMED.STOREYS              0.036686254  0.137316009
## BUILDING.CLEANLINESS           5.970127080  6.527029360
## BUILDING.EXTERIOR              3.466863694  3.971720706
## COMMON.AREA.PESTS              4.579156650  5.674131231
```

```
#
par(mfrow = c(1, 3))

#Residual vs fitted
plot(model, which = 1)

#QQ plot
plot(model, which = 2)

#Standardized residual vs fitted
plot(fitted(model),
     rstandard(model),
     main = "Standardized Residual vs Fitted",
     xlab = "Fitted values",
     ylab = "Standardized residuals")

abline(h = 0, col = "red")
```




```
par(mfrow = c(1, 1))

hist(
  rstandard(model),
  main = "Histogram of Standardized Residuals",
  xlab = "Standardized Residuals",
  col = "lightgray",
  border = "white"
)
```




```
numeric_vars <- clean_data[, c(
  "YEAR.BUILT",
  "CONFIRMED.UNITS",
  "CONFIRMED.STOREYS",
  "BUILDING.CLEANLINESS",
  "BUILDING.EXTERIOR"
)]
```

Preliminary Results

Several predictors were significantly associated with apartment building evaluation scores.

Building cleanliness, building exterior condition, and common area pests showed strong positive associations with current building evaluation scores.

The model explained approximately 51% of the variation in apartment building evaluation scores.